11.7.1 Decomposition method

Sometimes it is very difficult to integrate the given function directly. But it can be integrated after decomposing it into a sum or difference of number of functions whose integrals are already known.

For example $\left(1-x^3\right)^2$, $\frac{x^2-x+1}{x^3}$, $\cos 5x \sin 3x$, $\cos^3 x$, $\frac{e^{2x}-1}{e^x}$, do not have direct formulae to integrate. But these functions can be decomposed into a sum or difference of functions, whose individual integrals are known. In most of the cases the given integrand will be any one of the algebraic, trigonometric or exponential forms, and sometimes combinations of these functions.

Example 11.15

Integrate the following with respect to x:

(i) $\left(1-x^3\right)^2$ (ii) $\frac{x^2-x+1}{x^3}$

Solution

(i) $$\int \left(1-x^3\right)^2 dx = \int (1-2x^3+x^6)dx$$
$$= \int dx - 2\int x^3 dx + \int x^6 dx$$
$$= x - \frac{x^4}{2} + \frac{x^7}{7} + c.$$

(ii) $$\int \frac{x^2-x+1}{x^3} dx = \int \left(\frac{x^2}{x^3} - \frac{x}{x^3} + \frac{1}{x^3}\right) dx$$
$$= \int \frac{1}{x} dx - \int \frac{1}{x^2} dx + \int \frac{1}{x^3} dx.$$
$$\int \frac{x^2-x+1}{x^3} dx = \log|x| + \frac{1}{x} - \frac{1}{2x^2} + c.$$

Example 11.16

Integrate the following with respect to x:

(i) $\cos 5x \sin 3x$ (ii) $\cos^3 x$.

Solution

(i) $$\int \cos 5x \sin 3x \, dx = \frac{1}{2}\int 2\cos 5x \sin 3x \, dx$$
$$= \frac{1}{2}\int (\sin 8x - \sin 2x) dx$$
$$\int \cos 5x \sin 3x \, dx = \frac{1}{2}\left(-\frac{\cos 8x}{8} + \frac{\cos 2x}{2}\right) + c.$$

(ii) $$\int \cos^3 x \, dx = \frac{1}{4}\int (3\cos x + \cos 3x) dx$$
$$= \frac{1}{4}\left(3\sin x + \frac{\sin 3x}{3}\right) + c$$

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